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Structural Components of Alginic Acid. II. The Crystalline Structure of Poly-α-L-Guluronic Acid. Results of X-Ray Diffraction and Polarized Infrared Studies

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Synopsis

A structural investigation of the marine algal polysaccharide poly-α-L-guluronic acid is described. The molecular chains consist of 1 → 4 diaxially linked L-guluronic acid residues in the 1C chair conformation and are stabilized in a twofold helix conformation by an intra-molecular O(2)H...O(6)D hydrogen-bond. The X-ray fiber diffraction photograph has been indexed to an orthorhombic unit cell in which $a = 8.6$ Å, b (fiber axis) = 8.7 Å, $c = 10.7$ Å.

A structure corresponding to the space group $P2_12_12_1$ is proposed, in which all intermolecular hydrogen bonds interact with water molecules and in which all oxygen atoms except for the inaccessible bridge oxygens are involved. The relationship between the shape and structure of the polyguluronic acid molecule and its biological function is discussed.

INTRODUCTION

The crystalline structure of poly-α-L-guluronic acid was given in outline in a previous publication.[1] It was established that the monosaccharide units are in the 1C chair conformation and arranged to form a 1 → 4 diaxially linked polysaccharide chain. The molecular chains are in a twofold helical conformation with a fiber repeat of 8.7 Å, and are stabilized by an intrachain hydrogen bond between O(2)H of one residue and O(6)D of the next residue in the chain (see Fig. 1). The conformation is therefore more buckled than the flat ribbon-like structure associated with the 1 → 4 diequatorially linked hexosans. The present contribution is concerned with the detailed packing of the molecular chains and the precise nature of the inter- and intra-molecular hydrogen-bond scheme. Good crystalline order necessitates one water molecule per sugar residue and dehydration results in a uniplanar contraction of 1 Å in the a dimension[2] (perpendicular to the chain axis).

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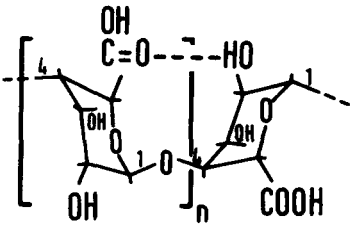


Fig. 1. Repeating unit of polyguluronic acid. The glucosidic linkage is 1a → 4a and the intra-chain hydrogen bond between O(2) and O(6)D is shown dotted.

MATERIALS AND METHODS

Alginic acid enriched in polyguluronic acid was prepared from fresh stipes of *Lamineria hyperborea* collected from Anglesey, Wales. Samples for X-ray diffraction and infrared experiments were prepared and examined as described in Part I.

Monomer coordinates in the 1C chair conformation were again derived from the data of Ramachandran et al.[3] and Nahringbauer.[4] The coordinates of the atoms in the chain were derived from the monomer coordinates using a C(1)–O–C(4′) angle of 117.5° and torsional angles φ,ψ = 13°, −14°, respectively, which correspond to a twofold axis along the chain direction and a projected axial repeat of 8.7/2 = 4.35 Å.

The conventions used for the origins and senses of rotation of the torsional angles are the same as those used in the polymannuronic structure investigation.

RESULTS

X-Ray Diffraction and Model Building

The X-ray fiber diffraction photograph shown in Figure 2 has been indexed to an orthorhombic unit cell with a = 8.6 ± 0.1 Å, b (fiber axis) = 8.7 ± 0.1 Å, c = 10.7 ± 0.1 Å. The limited number of reflections makes unequivocable space group assignment difficult but the extinction of (h00), (0k0) and (00l) reflections for h, k, and l = 1 and 3 suggest three sets of mutually perpendicular axes. The initial choice of the space group $P2_12_12_1$ was made on available evidence.

The unit-cell dimensions varied with humidity and on intensive drying a poorer diffraction pattern was obtained with dimensions a = 7.7 ± 0.1 Å, b (fiber axis) = 8.7 ± 0.1 Å, and c = 10.6 ± 0.1 Å. The measured density of polyguluronic acid is 1.59 g cm^{-3} which is in good agreement with the density (1.60 g cm^{-3}) calculated with one water molecule per residue (cf. the value of 1.45 g cm^{-3} was obtained for the unhydrated molecule).

The preliminary structure published by Atkins et al.[1] was used as a starting point for the present rigid body refinement, performed on the comparison between observed and calculated structure factors. The water molecules were positioned so as to achieve a sterically permissible

TABLE I
Fractional Atomic Coordinates of Poly-α-L-guluronic Acid

Atom	*X*	*Y*	*Z*
C(1)	0.103	0.282	0.314
C(2)	0.004	0.176	0.393
C(3)	−0.138	0.128	0.319
C(4)	−0.094	0.061	0.194
C(5)	0.016	0.170	0.125
C(6)	0.077	0.105	0.002
O(2)	−0.038	0.247	0.505
O(3)	−0.234	0.258	0.300
O(4)	−0.018	−0.082	0.213
O(5)	0.144	0.208	0.201
O(6)S	0.003	0.156	−0.097
O(6)D	0.185	0.014	−0.006
O(water)	−0.037	0.378	0.745

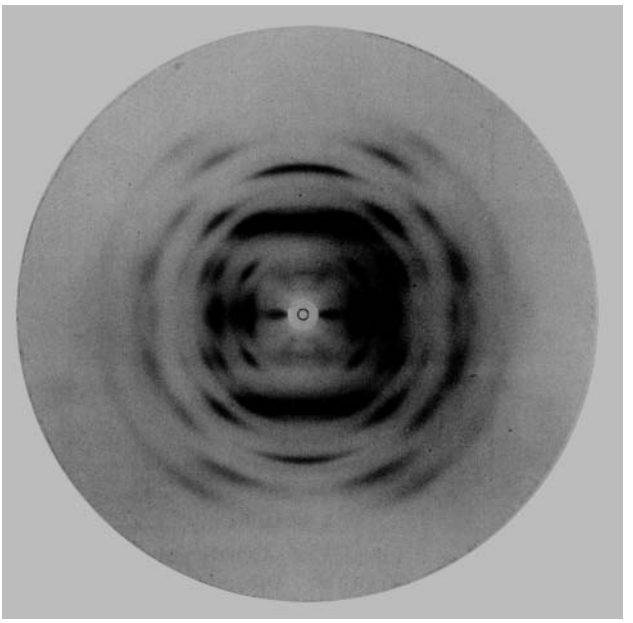


Fig. 2. X-ray diffraction fiber photograph obtained from polyguluronic acid. The pattern indexes on an orthorhombic unit cell with the fiber axis = 8.7 ± 0.1 Å.

hydrogen-bonding scheme and their inclusion resulted in an appreciable decrease in the residual index. Each water molecule has a nearly tetragonal environment of neighbors to provide hydrogen bonds in the range 2.8–2.9 Å and link three chains together (see Fig. 3).

In contrast to the chains themselves the (COOH) groups are relatively more rigidly defined due to intra-molecular contacts and Figure 4 shows that only a narrow range of rotations (χ = 90–120°) is sterically permissible.

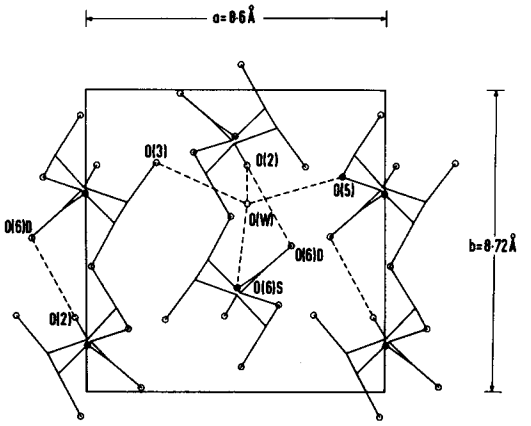


Fig. 3. *ab* projection of polyguluronic acid. The hydrogen-bonds are shown dashed. Only one water molecule, represented by O(W), showing its tetrahedral environment is included to avoid confusion.

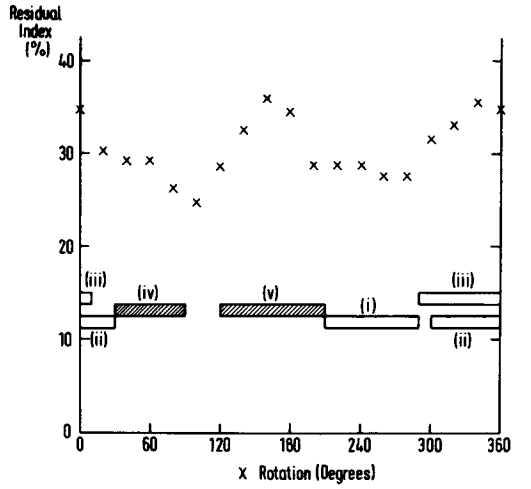


Fig. 4. Residual index R as a function of –COOH group rotation χ. Also intramolecular short contacts between (i) O(6)S...O(5); (ii) O(6)S...O(2′); (iii) O(6)S...O(4); (iv) O(6)D...O(5); (v) O(6)D...O(2′). Cross hatched boxes refer to O(6)D short contact region and unshaded boxes refer to O(6)S short contact region.

In calculating contact parameters it was considered that an HO-....O distance less than 2.4 Å constituted a short contact so that no hydrogen bond was formed.

Examination of both scale wire models and space-filling models indicated that two *intra*-molecular hydrogen bonds were possible when the mean position of $\chi = 100°$ was chosen; O(2)-...O(6)D = 2.65 Å and O(3)-...O(6)D = 3.3 Å. It was argued that the former hydrogen bond would be more probable.

TABLE II
Observed and Calculated Structure Factors

Miller index	$F_{obs.}$	$F_{calc.}$
101	10.0	11.0
002	6.6	7.4
102	3.1	3.1
200	15.4	15.3
201	4.3	7.4
004	6.6	8.1
203	3.1	3.8
301	3.7	4.6
302	3.8	2.8
104	5.6	4.1
303	4.5	2.6
204	1.1	0.6
010	*	0
111	4.1	7.0
012	1.9	0.5
112	5.6	6.8
211	5.3	4.8
113	3.4	1.3
014	6.1	4.9
213	5.2	4.2
311	7.6	6.1
312	2.4	2.6
114	3.0	3.4
313	4.5	4.8
214	2.6	2.8
020	12.1	9.5
021	7.1	6.0
120	8.3	6.8
121	6.5	3.4
122	7.4	7.8
123	0.6	0.9
222	6.5	9.3
322	2.9	5.9
124	3.3	6.6
030	0	0.0
031	5.6	6.4
131	2.4	1.9
132	6.0	3.4
230	4.3	2.4
133	1.9	2.0
232	6.3	6.7

* See Atkins and Parker.[11]

A plot of residual index R against χ is shown in Figure 4 and indicates that the steric considerations and hydrogen bonding possibilities favor $\chi = 100°$.

The coordinates of the proposed structure are given in Table I and the observed and calculated structure factors in Table II. Projections of the

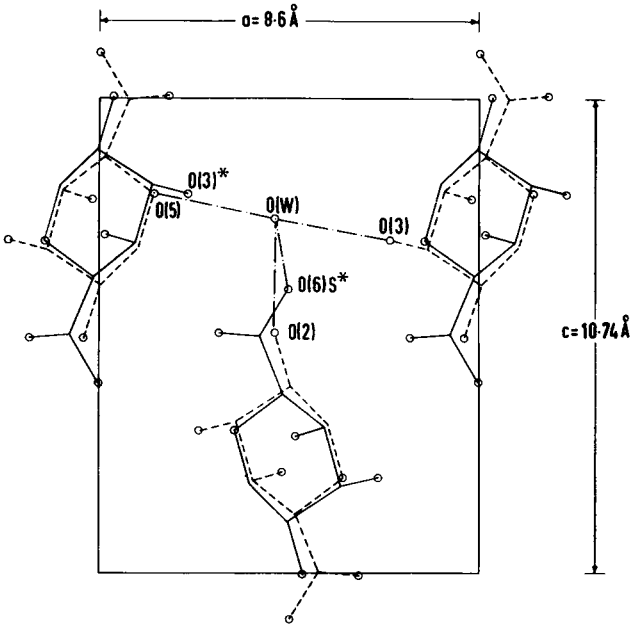


Fig. 5. Equatorial projection for polyguluronic acid. The hydrogen bonds are represented by ·-·-·. The asterisk denotes two atoms which may also be hydrogen bonded. The dotted lines represent the next residue in the chains.

structure in the *a,b* plane is shown in Figure 3 and in the *a,c* plane in Figure 5.

Infrared Spectroscopy

Hydroxyl Stretching Region. The absorption is diffuse and without any marked peaks, even, surprisingly, after partial deuteration (see Fig. 6). This may be attributed to a lack of precision in hydrogen-bond formation in the structure and in consequence little information of structural value is to be obtained.

There is, however, a suggestion in both natural and deuterated samples of a very weak hydrogen bond at about 3520 cm^{-1} (corresponding to 2.91 Å) which has parallel dichroism, and of a weak hydrogen bond at 3450 cm^{-1} (2.84 Å) which is perpendicular. The absorption continues out to 2000 cm^{-1} and so we may expect that this is due to carboxyl proton. If so it tends to be perpendicular to the fiber axis.

Carbonyl Stretching Band. A single perpendicular band is observed centered at 1734 cm^{-1} with a dichroic ratio of 1:1.5 corresponding to $\theta > 60°$. There is no indication of splitting (of the kind found in polymannuronic acid) but the very large bandwidth of 61 cm^{-1} suggests that the carbonyl group does not have a well-defined environment throughout the greater part of this specimen. The frequency of the maximum is significantly higher than that occurring in crystalline polymannuronic acid.

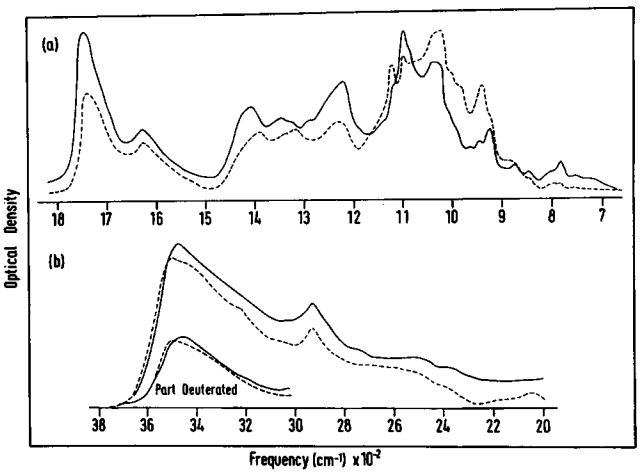


Fig. 6. Infrared spectrum obtained from polyguluronic acid. (a) Polarized infrared absorption in the frequency range 1800–1700 cm^{-1}. (b) Polarized infrared absorption in the frequency range 3800–2000 cm^{-1}. The effect of attempted deuteration is shown. ---$A_{||}$, ——$A_{\perp}$.

Absorption 1200–700 cm^{-1}

Despite the diffuseness of the hydroxyl and carbonyl bands the skeletal ring modes based upon the –C–O– grouping which absorb between 1200–700 cm^{-1} indicate that the basis of a unique structure is in fact present. The pattern is very different from that of polymannuronic acid; the bands are fairly sharp and there is moderate dichroism. At present no absorptions in the region can be assigned to vibrational modes but the overall pattern may be regarded as a characteristic of the polyguluronic acid structure.

DISCUSSION

The good agreement between observed and calculated structure factors supports the earlier proposal that in polyguluronic acid, the 1 → 4 linked monosaccharide units are in the 1C chair form and are diaxially linked.[5]

The polymer chain has a twofold repeat but as a result of the 1a → 4a linkage the conformation is buckled compared to the polymannuronic acid structure. There are two positions in the conformational map which satisfy the requirements of a twofold helix, with an axial repeat of 4.35 Å, and these conformations are far from the allowed area when the monosaccharide is in the C1 chair form or in either chair form but diequatorially 1 → 4 linked. Like polymannuronic acid only 3.3% of possible backbone conformations fall in the allowed region and in the present case the allowed positions of the COOH group are considerably more restricted than in polymannuronic acid. The chain conformation and side (COOH) group position imply that a strong hydrogen bond (~2.7 Å) can be formed be-

tween O(2)H and O(6)D. The infrared results do not conclusively confirm such a bond but the spectrum does show evidence for a parallel hydrogen bond. The high frequency of the band (3500 cm^{-1}) would indicate a rather weaker hydrogen bond longer than 2.7 Å. It is possible however that the frequency is altered because O(2) is probably involved in hydrogen bond formation with water molecules.

It is important to consider why the crystalline regions of polyguluronic acid take up water. First, it seems that the molecule is a poor space-filler since the zigzag shape leaves a large cleft which is largely hydrophobic and unfilled. Secondly, axial hydroxyl groups appear, in general, to have a lower hydrogen-bonding capability. Thirdly, as discussed above, the molecule has very little flexibility (see also Ref. 6) and therefore a low inter-molecular hydrogen-bonding potential. Thus, water plays a very important binding role; it can be seen (Fig. 5) that all of the inter-molecular links involve water molecules as in the structure of β, 1 → 4-xylan hydrate[7] and the ratio of one water molecule to one residue of guluronic acid is not exceeded since all oxygen atoms, other than the bridge oxygen which is inaccessible, are already involved in a hydrogen-bonding scheme. Finally, it is known that alginates with a high content of guluronic acid have great affinities for divalent cations. Furthermore, guluronate salts retain a two fold helical conformation[8] unlike mannuronates, hyaluronates,[9] and certain of the soda celluloses,[10] although a lefthanded three-fold conformation would be sterically permissible.

CONCLUSION

The structure factor calculations support the hypothesis that the L-guluronic acid residue adopts the 1C chair conformation and the results indicate that one aspect of the inflexibility of polyguluronic acid is the very limited number of permissible side (COOH) group conformations.

Whereas, polymannuronic acid has a closely packed sheet-like structure, the more buckled conformation adopted by polyguluronic acid leaves sites for the inclusion of water molecules. The conformations of both molecules are stabilized by intra-molecular hydrogen bonds but, since a carboxyl oxygen atom carries a considerably greater partial negative charge than a ring oxygen the strength of the bond in polyguluronic acid is presumably greater, thus contributing to its greater rigidification.

Guluronate salts which adopt a similar backbone conformation will also be restricted in side group rotation and it seems very probable that metallic cations may occupy the same sites as the water molecules in the acid form.

It is clear that in a biological environment the shape, flexibility and other properties of the polymannuronic/polyguluronic acid system can be tailored to specific requirements. Thus, the blending of the properties of both the homopolymers and the alternating copolymer together with their salt forms produces a wide spectrum of functions.

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